

RELAYING CREDENTIALS HAS NEVER BEEN EASIER:

**HOW TO EASILY BYPASS THE LATEST NTLM RELAY
MITIGATIONS**



**Marina Simakov
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ABOUT US

Marina Simakov (@simakov_marina)

- Senior Security Researcher @Preempt
- M.Sc. in computer science, with several published articles, with a main area of expertise in graph theory
- Previously worked as a Security Researcher @Microsoft
- Spoke at various security conferences such as Black Hat, Blue Hat IL and DefCon

Yaron Zinar (@YaronZi)

- Senior Security Researcher Lead @Preempt
- M.Sc. in Computer Science with a focus on statistical analysis
- Spent over 12 years at leading companies such as Google and Microsoft
- Among his team latest finding are CVE-2017-8563, CVE-2018-0886, CVE-2019-1040 and CVE-2019-1019

AGENDA

1. Introduction:

- Common attacks on Active Directory
- NTLM
 - Design weaknesses
 - NTLM Relay
 - Offered mitigations

2. Known Vulnerabilities

- LDAPS Relay
- CVE-2015-0005

3. New vulnerabilities

- Your session key is my session key
- Drop the MIC
- EPA bypass
- Attacking AD FS
 - External lockout bypass
 - Reverse-Kerberoasting

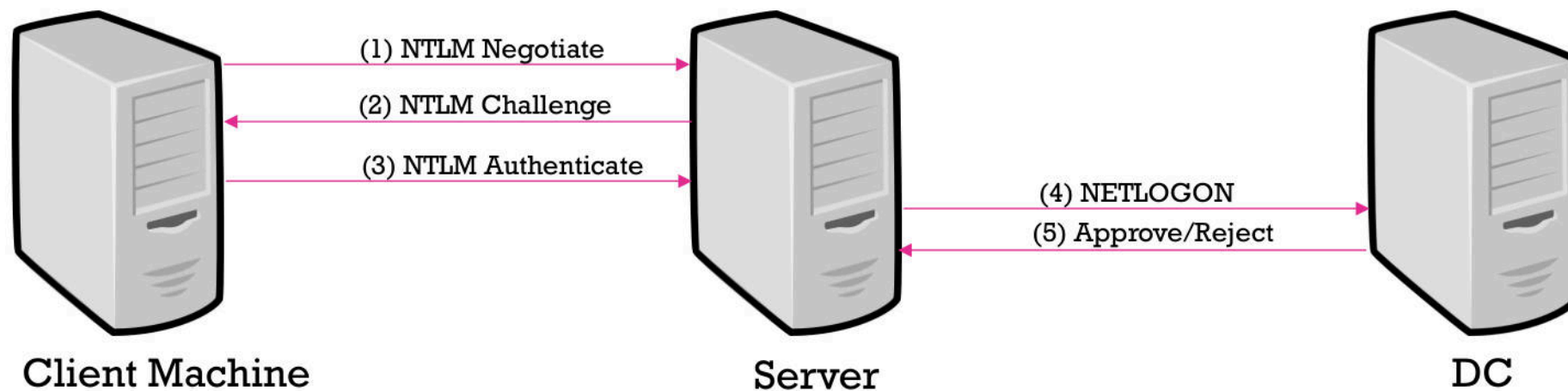
4. Takeaways



INTRODUCTION: ACTIVE DIRECTORY

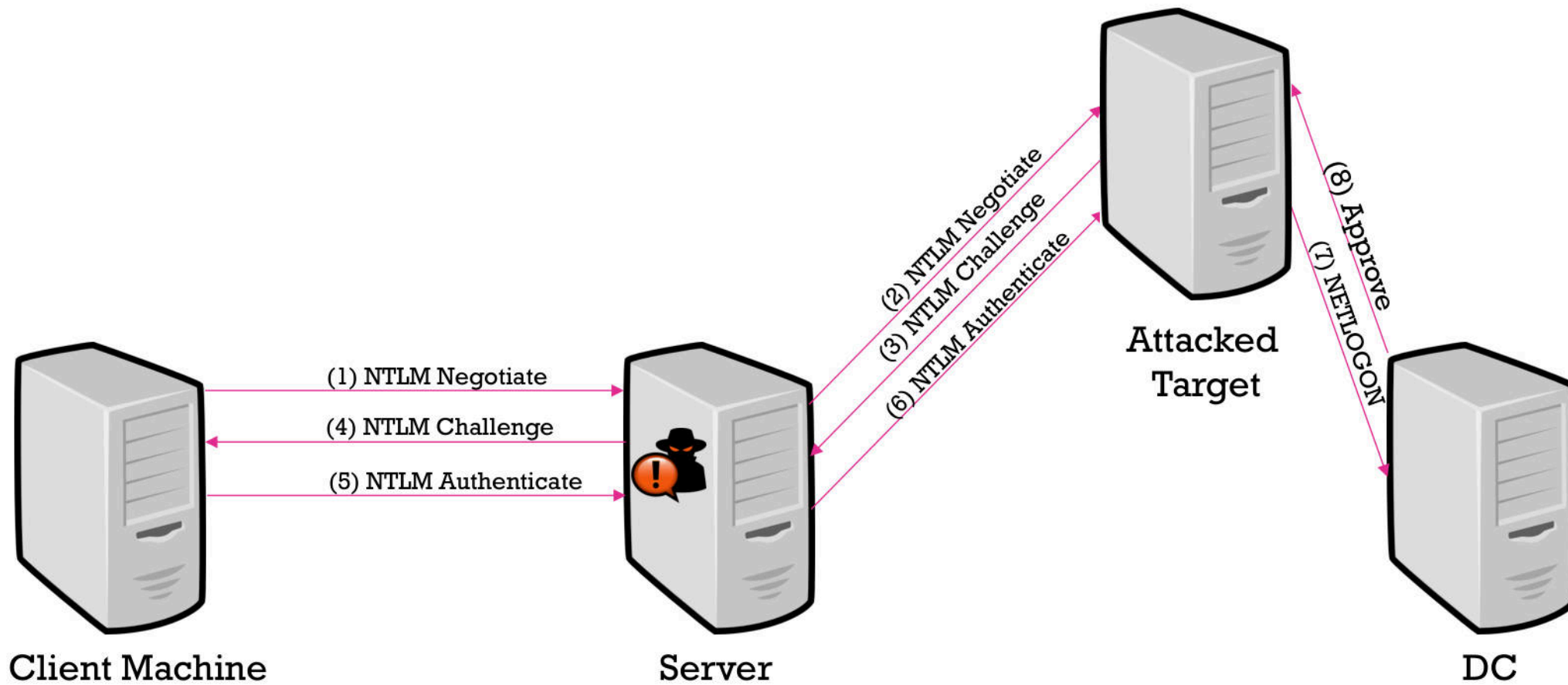
- Main secrets storage of the domain
 - Stores password hashes of all accounts
 - In charge of authenticating accounts against domain resources
- Authentication protocols
 - LDAP
 - NTLM
 - Kerberos
- Common attacks
 - Golden & Silver Ticket
 - Forged PAC
 - PTT
 - PTH
 - NTLM Relay

NTLM



Authentication is *not* bound to the session!

NTLM RELAY





NTLM RELAY: MITIGATIONS

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NTLM RELAY: MITIGATIONS

- Mitigations:
 - SMB Signing
 - LDAP Signing
 - EPA (Enhanced Protection for Authentication)
 - LDAPS channel binding
 - Server SPN target name validation
 - Hardened UNC Paths

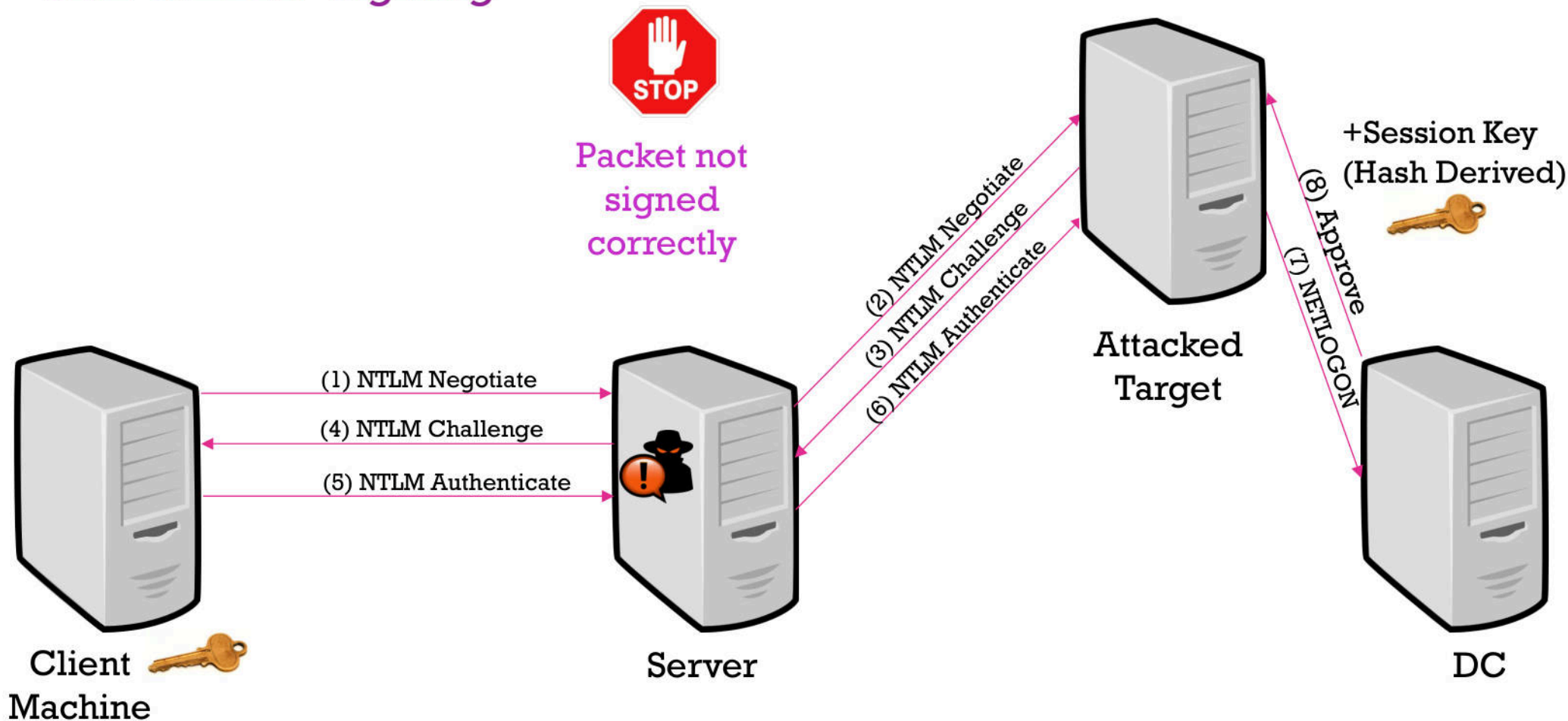
NTLM RELAY: MITIGATIONS

- SMB & LDAP signing

- After the authentication, all communication between client and server will be signed
- The signing key is derived from the authenticating account's password hash
- The client calculates the session key by itself
- The server receives the session key from the DC in the NETLOGON response
- An attacker with relay capabilities has no way of retrieving the session key

NTLM RELAY: MITIGATIONS

- SMB & LDAP signing

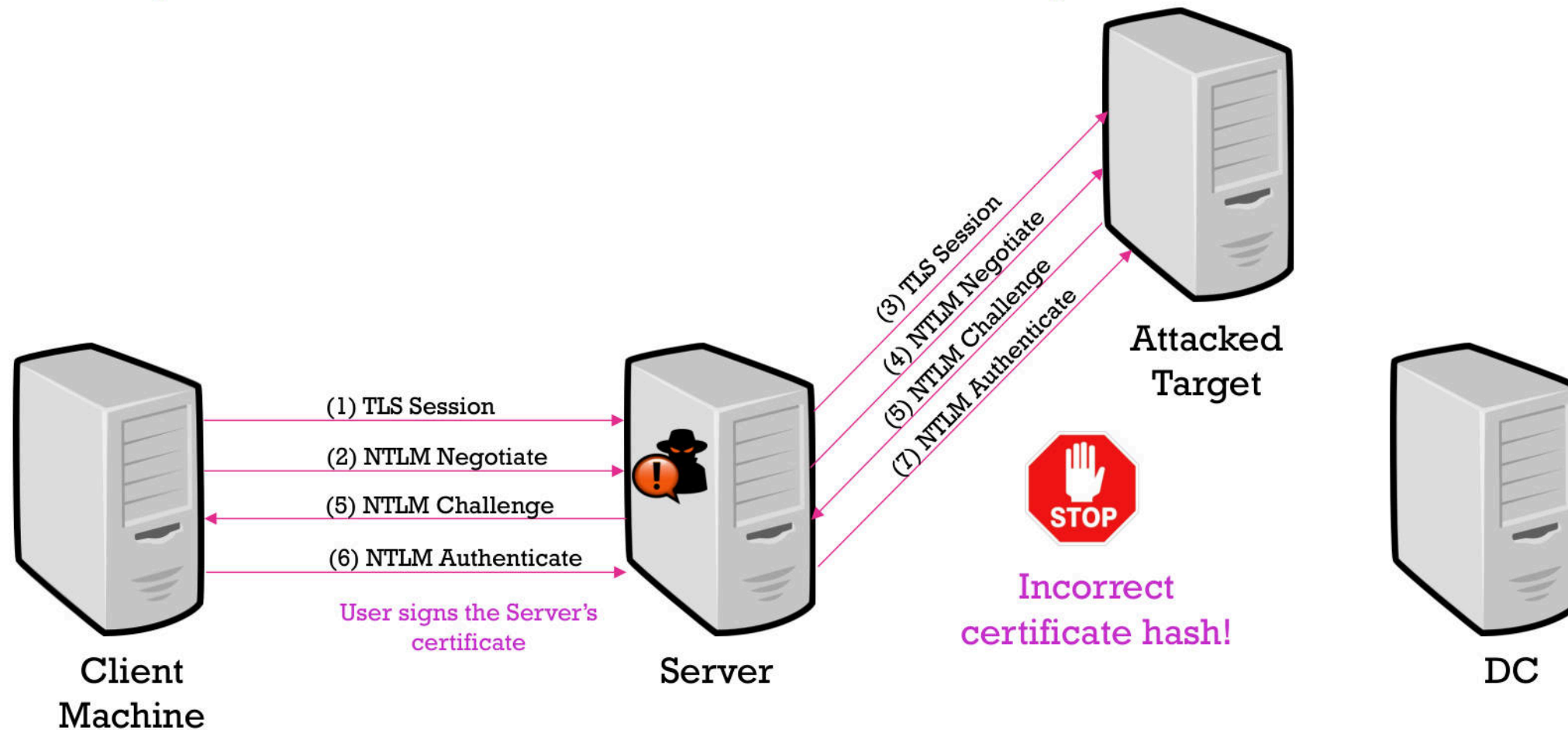


NTLM RELAY: MITIGATIONS

- EPA (Enhanced Protection for Authentication)
 - RFC 5056
 - Binds the NTLM authentication to the secure channel over which the authentication occurs
 - The final NTLM authentication packet contains a hash of the target service's certificate, signed with the user's password hash
 - An attacker with relay capabilities is using a different certificate than the attacked target, hence the client will respond with an incompatible certificate hash value

NTLM RELAY: MITIGATIONS

- EPA (Enhanced Protection for Authentication)





NTLM RELAY: KNOWN VULNERABILITIES

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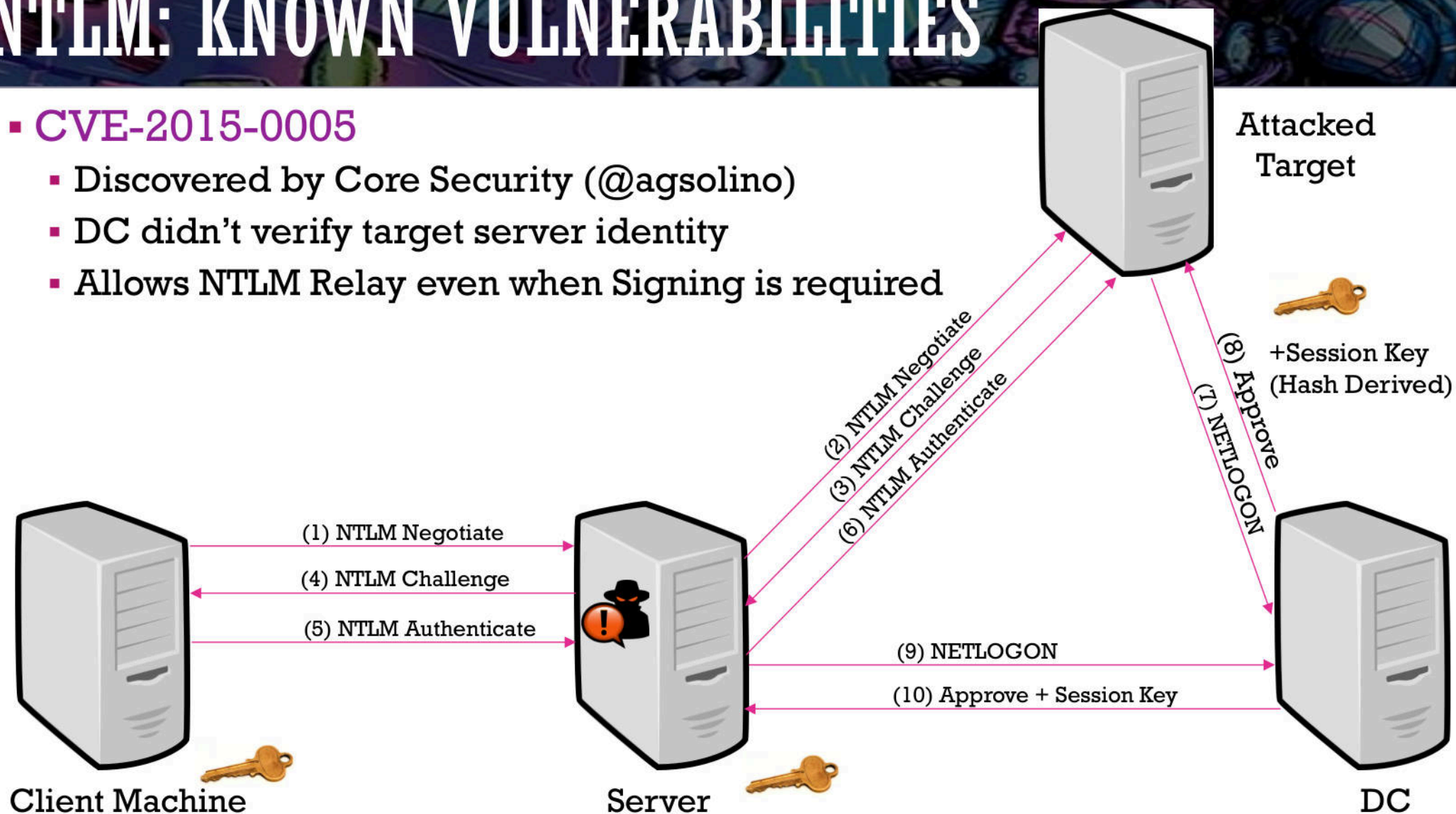
NTLM: KNOWN VULNERABILITIES

- **LDAPS Relay (CVE-2017-8563)**
 - Discovered by Preempt in 2017
<https://blog.preempt.com/new-ldap-rdp-relay-vulnerabilities-in-ntlm>
 - Group Policy Object (GPO) - *“Domain Controller: LDAP server signing requirements”*
 - Requires LDAP sessions to be signed **OR**
 - Requires session to be encrypted via TLS (LDAPS)
- TLS does not protect from credential forwarding!

NTLM: KNOWN VULNERABILITIES

- **CVE-2015-0005**

- Discovered by Core Security (@agsolino)
- DC didn't verify target server identity
- Allows NTLM Relay even when Signing is required



NTLM: KNOWN VULNERABILITIES

- CVE-2015-0005

- NTLM Challenge message:

- Contains identifying information about the target computer

```
NTLM Secure Service Provider
NTLMSSP identifier: NTLMSSP
NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)
  Target Name: PREEMPT
  Negotiate Flags: 0x02898205, Negotiate Version, Negotiate Target Info,
  NTLM Server Challenge: 5254321a3ca3b35b
  Reserved: 0000000000000000
  Target Info
    Length: 164
    Maxlen: 164
    Offset: 76
    Attribute: NetBIOS domain name: PREEMPT
    Attribute: NetBIOS computer name: TEST-01
    Attribute: DNS domain name: preempt
    Attribute: DNS computer name: TEST-01.preempt
    Attribute: DNS tree name: preempt
    Attribute: Timestamp
    Attribute: End of list
  Version 6.3 (Build 9600); NTLM Current Revision 15
```

Attacked Target

NTLM: KNOWN VULNERABILITIES

- CVE-2015-0005

- NTLM Authenticate message:

- User calculates HMAC_MD5 based on the challenge message using his NT Hash

```
NTLMv2 Response: 6c1da1bba6a09b2f637a7a18b20eb1650101000000000000...
```

```
NTProofStr: 6c1da1bba6a09b2f637a7a18b20eb165
```

```
Response Version: 1
```

```
Hi Response Version: 1
```

```
Z: 000000000000
```

```
Time: May 28, 2019 08:21:41.061147500 UTC
```

```
NTLMv2 Client Challenge: 2d30979d36e171b5
```

```
Z: 00000000
```

```
▷ Attribute: NetBIOS domain name: PREEMPT
```

```
▷ Attribute: NetBIOS computer name: TEST-01
```

```
▷ Attribute: DNS domain name: preempt
```

```
▷ Attribute: DNS computer name: TEST-01.preempt
```

```
▷ Attribute: DNS tree name: preempt
```

```
▷ Attribute: Timestamp
```

```
▷ Attribute: Flags
```

```
▷ Attribute: Restrictions
```

```
▷ Attribute: Channel Bindings
```

```
▷ Attribute: Target Name: cifs/10.1.1.1
```

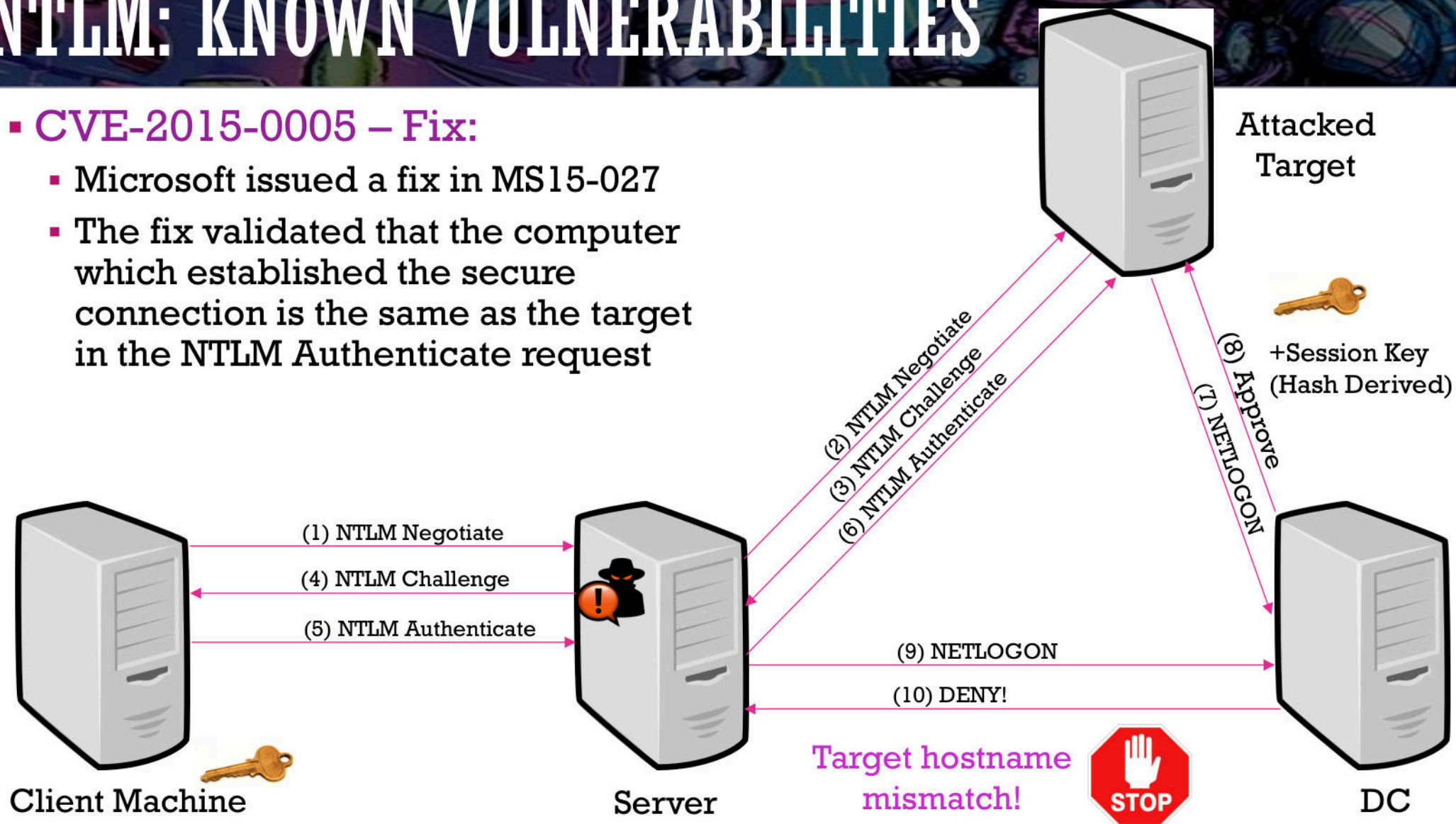
```
▷ Attribute: End of list
```

Attacked Target

NTLM: KNOWN VULNERABILITIES

- **CVE-2015-0005 – Fix:**

- Microsoft issued a fix in MS15-027
- The fix validated that the computer which established the secure connection is the same as the target in the NTLM Authenticate request





NTLM RELAY: NEW VULNERABILITIES

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NTLM: NEW VULNERABILITIES

- **Your session key is my session key**
 - Retrieve the session key for any NTLM authentication
 - Bypasses the MS15-027 fix
- **Drop the MIC**
 - Modify session requirements (such as signing)
 - Overcome the MIC protection
- **EPA bypass**
 - Relay authentication to servers which require EPA
 - Modify packets to bypass the EPA protection
- **Attacking AD-FS**
 - External lockout policy bypass
 - Reverse-Kerberoasting



**ARE YOU TELLING ME
ALL NTLM RELAY MITIGATIONS**

ARE COMPLETELY USELESS?

imgflip.com



**YOUR SESSION KEY IS MY
SESSION KEY**

NTLM: NEW VULNERABILITIES

- Your session key is my session key
 - MS15-027 fix validates target NetBIOS name
 - But what if the target NetBIOS name field is missing?



Original challenge:

```
NTLM Secure Service Provider
NTLMSSP identifier: NTLMSSP
NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)
  Target Name: PREEMPT
  Negotiate Flags: 0x02898205, Negotiate Version, Negotiate
  NTLM Server Challenge: 5254321a3ca3b35b
  Reserved: 0000000000000000
  Target Info
    Length: 164
    Maxlen: 164
    Offset: 76
    Attribute: NetBIOS domain name: PREEMPT
    Attribute: NetBIOS computer name: TEST-01
    Attribute: DNS domain name: preempt
    Attribute: DNS computer name: TEST-01.preempt
    Attribute: DNS tree name: preempt
    Attribute: Timestamp
    Attribute: End of list
  Version 6.3 (Build 9600); NTLM Current Revision 15
```

Modified challenge:

```
NTLM Secure Service Provider
NTLMSSP identifier: NTLMSSP
NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)
  Target Name: PREEMPT
  Negotiate Flags: 0x02898205, Negotiate Version, Negotiate
  NTLM Server Challenge: 5254321a3ca3b35b
  Reserved: 0000000000000000
  Target Info
    Length: 164
    Maxlen: 164
    Offset: 76
    Attribute: NetBIOS domain name: PREEMPT
    Attribute: DNS domain name: preempt
    Attribute: DNS computer name: TEST-01.preempt
    Attribute: DNS tree name: preempt
    Attribute: Timestamp
    Attribute: End of list
  Version 6.3 (Build 9600); NTLM Current Revision 15
```



NTLM: NEW VULNERABILITIES

- Your session key is my session key
 - The client responds with an NTLM_AUTHENTICATE message with target NetBIOS field missing
 - The NETLOGON message is sent without this field
 - The domain controller responds with a session key!

NTLM: NEW VULNERABILITIES

- **Your session key is my session key**
 - But what if the NTLM AUTHENTICATE message includes a **MIC**?
 - MIC: Message integrity for the NTLM NEGOTIATE, NTLM CHALLENGE, and NTLM AUTHENTICATE
 - MIC = HMAC_MD5(SessionKey, ConcatenationOf(NTLM_NEGOTIATE, NTLM_CHALLENGE, NTLM_AUTHENTICATE))

- NTLM Secure Service Provider

NTLMSSP identifier: NTLMSSP

NTLM Message Type: NTLMSSP AUTH (0x00000003)

```
> Lan Manager Response: 00000000000000000000000000000000000000000000000
```

LMv2 Client Challenge: 0000000000000000

```
> NTLM Response: 1336da946b1e967178af213a953bc69b0101000000000000...
```

```
> Domain name: PREEMPT
```

```
> User name: user01
```

```
> Host name: TEST-01
```

```
> Session Key: b694a2f88063a2fc0e8f122d33b90523
```

```
> Negotiate Flags: 0xe2888215, Negotiate 56, Negotiate Key Exchange, Negotiate 128,
```

```
> Version 10.0 (Build 17134); NTLM Current Revision 15
```

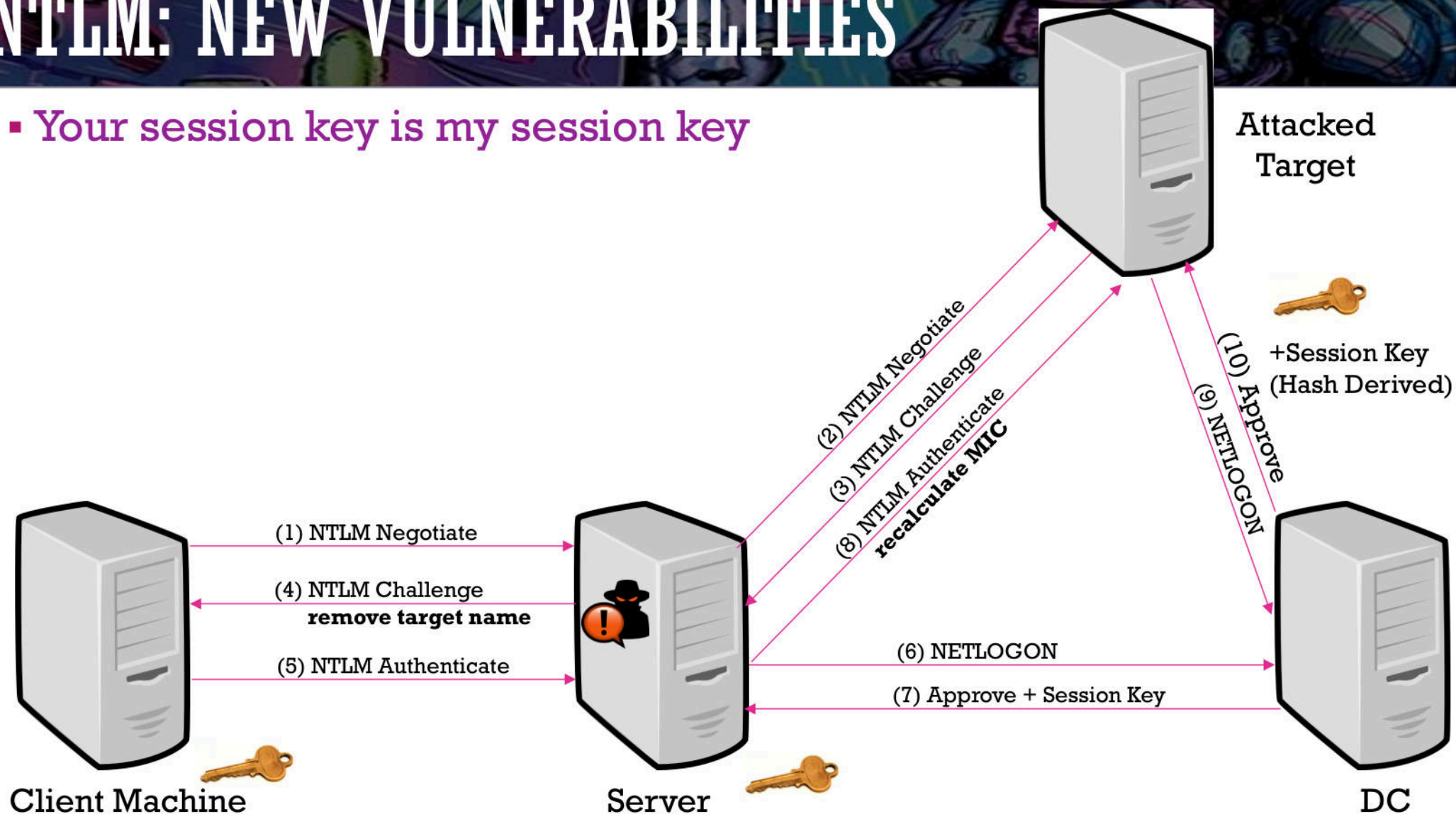
MIC: 7b7f086333cdd6d48a694c3c0cd2aa8d

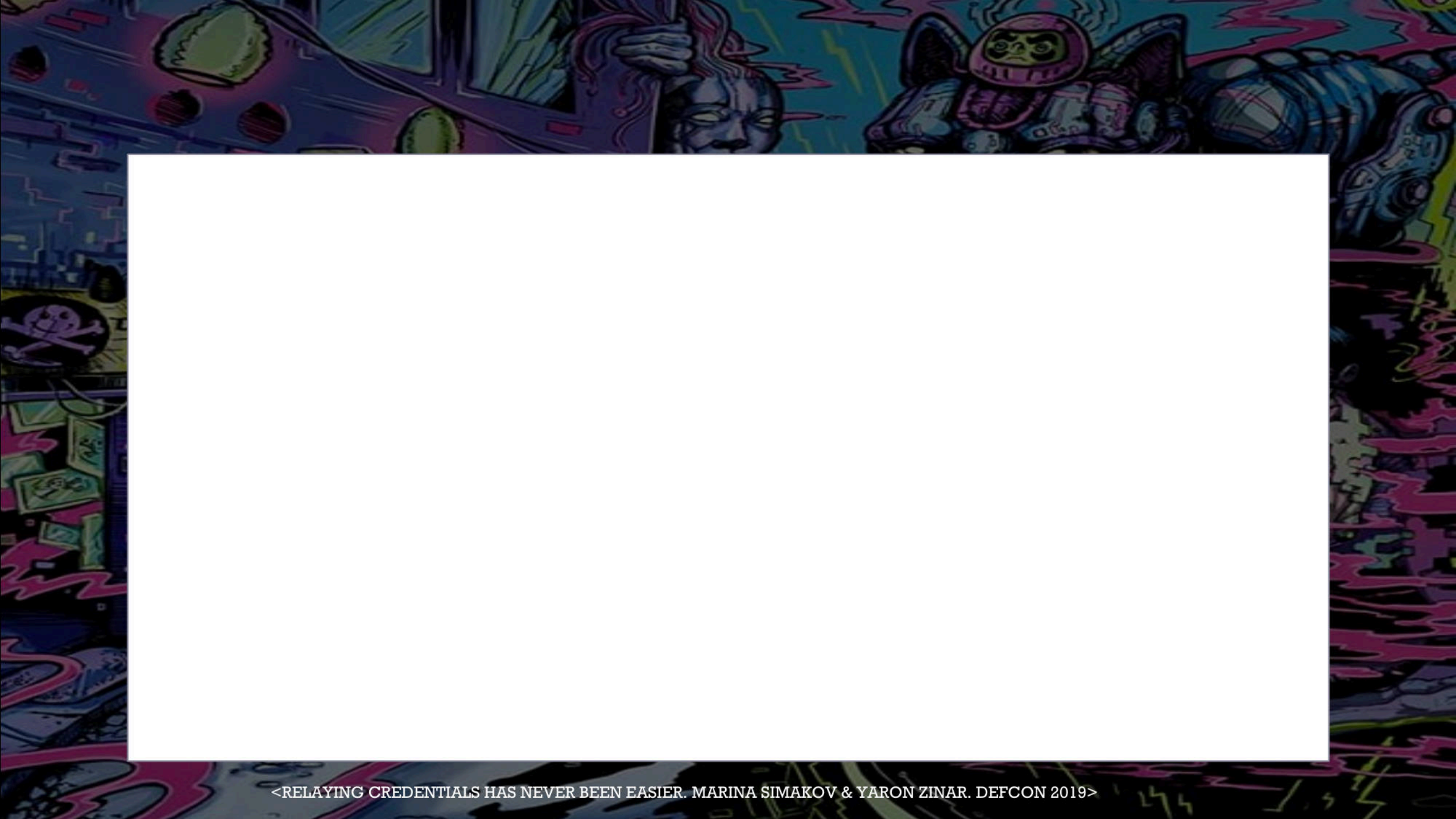
NTLM: NEW VULNERABILITIES

- Your session key is my session key
 - Overcoming the MIC problem:
 - By removing the target hostname we are able to retrieve the session key
 - We have all 3 NTLM messages
 - The client provides a MIC which is based on the modified NTLM_CHALLENGE message
 - We recalculate the MIC based on the original NTLM_CHALLENGE message

NTLM: NEW VULNERABILITIES

- Your session key is my session key





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SESSION KEYS



FOR EVERYONE

NTLM: NEW VULNERABILITIES

- Your session key is my session key – Fix:
 - Windows servers deny requests which do not include a target
- Issues:
 - NTLMv1
 - messages do not have av_pairs -> no target field
 - Such authentication requests remain vulnerable to the attack
 - Non-Windows targets are still vulnerable
 - Patching is not enough



DROP THE MIC

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NTLM: NEW VULNERABILITIES

- Drop the MIC

- MIC = HMAC_MD5(SessionKey, ConcatenationOf(NTLM_NEGOTIATE, NTLM_CHALLENGE, NTLM_AUTHENTICATE))

- If client & server negotiate session privacy/integrity, attackers cannot take over the session

- ▼ NTLM Secure Service Provider

- NTLMSSP identifier: NTLMSSP

- NTLM Message Type: NTLMSSP_NEGOTIATE (0x00000001)

- ▼ Negotiate Flags: 0xe2088297, Negotiate 56, Negotiate Key Exchange,

-
..... ..1 = Negotiate Sign: Set
.....

- Calling workstation domain: NULL

- Calling workstation name: NULL

- > Version 10.0 (Build 17134); NTLM Current Revision 15

- The MIC protects the NTLM negotiation from tampering

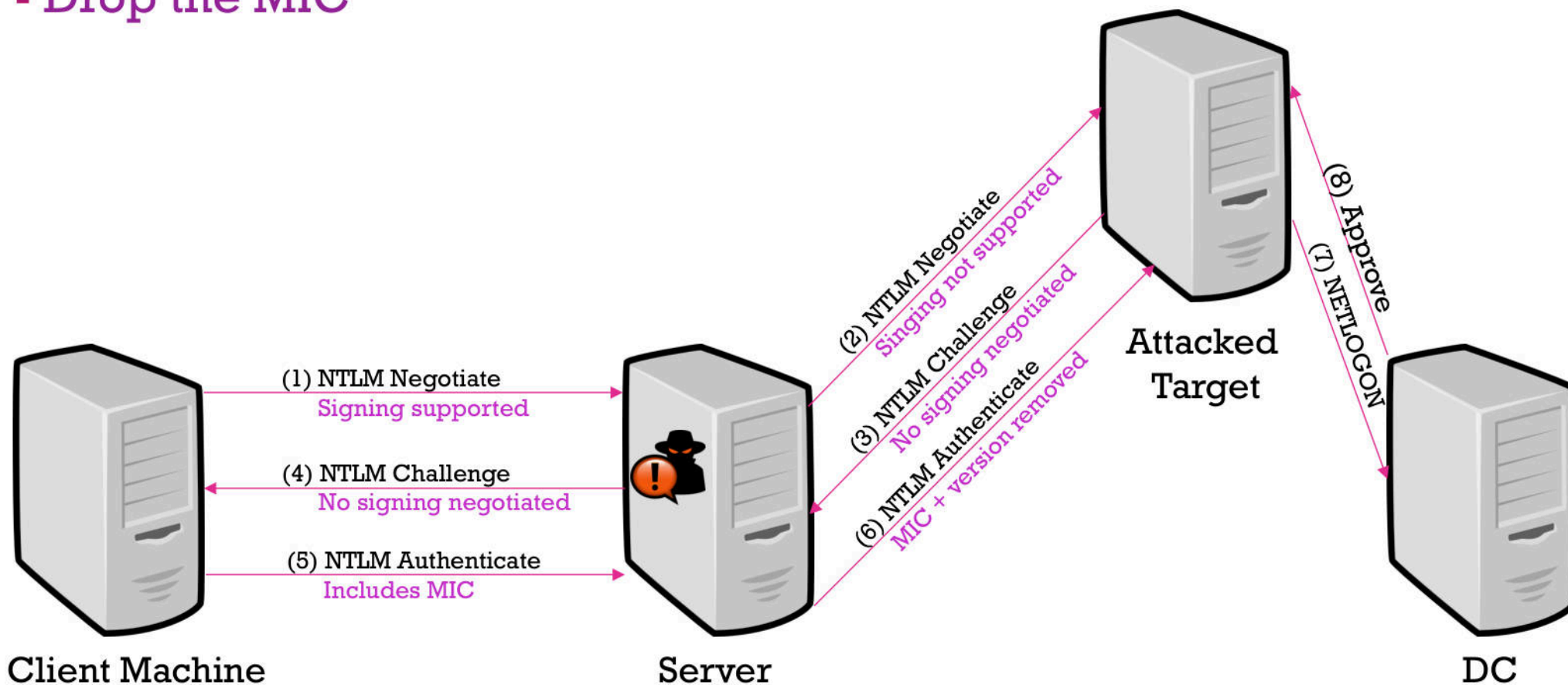
NTLM: NEW VULNERABILITIES

- Drop the MIC

- SMB clients turn on the signing negotiation flag by default & use a MIC
- It is not possible (or at least, not trivial) to relay SMB to another protocol which relies on this negotiation flag (in contrast to other protocols such as HTTP)
- How can we overcome this obstacle?
 - MIC can be modified only if the session key is known
 - Otherwise, it can be simply removed 😊
 - [In order to remove the MIC, the version needs to be removed as well, as well as some negotiation flags]
- Result: It is possible to tamper with any stage of the NTLM authentication flow when removing the MIC

NTLM: NEW VULNERABILITIES

- Drop the MIC



NTLM: NEW VULNERABILITIES

- Drop the MIC - Problem

- The MIC presence is notified in the `msvAvFlags` attribute in the NTLM authentication message
- `msvAvFlags` is signed with the user's password hash

MsvAvFlags 0x0006	A 32-bit value indicating server or client configuration. 0x00000001: Indicates to the client that the account authentication is constrained. 0x00000002: Indicates that the client is providing message integrity in the MIC field (section 2.2.1.3) in the AUTHENTICATE_MESSAGE.<14> 0x00000004: Indicates that the client is providing a target SPN generated from an untrusted source.<15>
----------------------	---

```
▷ Attribute: Timestamp
▷ Attribute: Flags
  NTLMV2 Response Item Type: Flags (0x0006)
  NTLMV2 Response Item Length: 4
  Flags: 0x00000002
▷ Attribute: Restrictions
▷ Attribute: Channel Bindings
▷ Attribute: Target Name: cifs/10.1.0.107
▷ Attribute: End of list
```

- Even if the corresponding bit is set, the target server does not verify that the MIC is indeed present



DROP THE MIC

Global
NEWS

NTLM: NEW VULNERABILITIES

- **MIC bypass - Fix:**

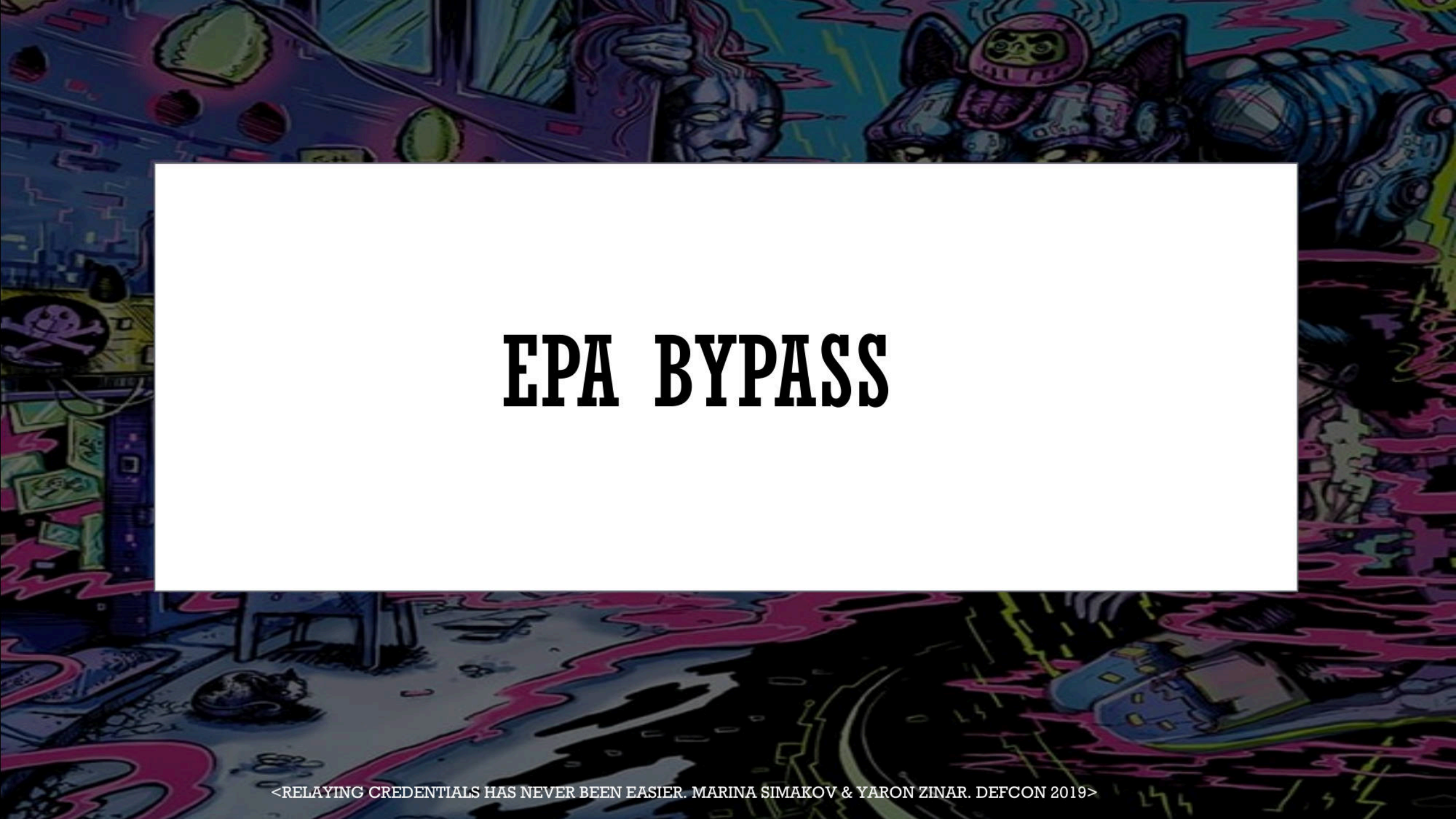
- If msvAvFlags indicate that a MIC is present, verify its presence.

- **Issues:**

- Some clients don't add a MIC by default (Firefox on Linux or MacOS)
 - These clients are still vulnerable to NTLM session tampering

- More serious issue:
CVE-2019-1166 –
Drop The MIC 2 ☺





EPA BYPASS

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NTLM: NEW VULNERABILITIES

- EPA (Enhanced Protection for Authentication) bypass

- EPA binds authentication packets to a secure TLS channel
- Adds a Channel Bindings field to the NTLM_AUTHENTICATE message based on the target server certificate
- Prevents attackers from relaying the authentication to another server
- Modification requires knowledge of the user's NT HASH

▼ NTLMv2 Response: 848ad4f1104a741871069e735d124a120101000000000000...

NTProofStr: 848ad4f1104a741871069e735d124a12

Response Version: 1

Hi Response Version: 1

Z: 000000000000

Time: May 30, 2019 11:04:16.356383400 UTC

NTLMv2 Client Challenge: e35869f876174a6f

Z: 00000000

> Attribute: NetBIOS domain name: PREEMPT

> Attribute: NetBIOS computer name: TEST-01

> Attribute: DNS domain name: preempt

> Attribute: DNS computer name: TEST-01.preempt

> Attribute: DNS tree name: preempt

> Attribute: Timestamp

> Attribute: Flags

> Attribute: Restrictions

▼ Attribute: Channel Bindings

NTLMV2 Response Item Type: Channel Bindings (0x000a)

NTLMV2 Response Item Length: 16

Channel Bindings: 26b0b57ea3af3852664834351af38549

> Attribute: Target Name: HTTP/10.1.1.1

> Attribute: End of list

NTLM: NEW VULNERABILITIES

- EPA (Enhanced Protection for Authentication) bypass
 - Servers protected by EPA:
 - AD-FS
 - OWA
 - LDAPS
 - Other HTTP servers (e.g. Sharepoint)
 - Unfortunately by default, EPA is disabled on all of the above servers
 - In most cases, these servers are vulnerable to much simpler attack vectors

NTLM: NEW VULNERABILITIES

- EPA (Enhanced Protection for Authentication) bypass

- Modifying the Channel Bindings in the NTLM_AUTHENTICATE message is not possible
- But what if we add a Channel Bindings field to the NTLM_CHALLENGE message before we send it to the client?

- ▼ NTLM Secure Service Provider

NTLMSSP identifier: NTLMSSP

NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)

> Target Name: PREEMPT

> Negotiate Flags: 0xe2898215, Negotiate 56, Negotiate Key Exchange, NTLM Server Challenge: cd755f40de40662d

Reserved: 0000000000000000

- ▼ Target Info

Length: 184

Maxlen: 184

Offset: 76

> Attribute: NetBIOS computer name: TEST-01

> Attribute: NetBIOS domain name: PREEMPT

> Attribute: DNS computer name: TEST-01.preempt

> Attribute: DNS domain name: preempt

> Attribute: DNS tree name: preempt

> Attribute: Timestamp

- ▼ Attribute: Channel Bindings

Target Info Item Type: Channel Bindings (0x000a)

Target Info Item Length: 16

Channel Bindings: 26b0b57ea3af3852664834351af38549

> Attribute: End of list

NTLM: NEW VULNERABILITIES

- EPA (Enhanced Protection for Authentication) bypass

- Client will add our crafted field to the NTLM_AUTHENTICATE message!
- Additional fields would be added to the message, including a second Channel Binding
- Server takes the first Channel Binding for verification

```
▼ NTLMv2 Response: b0eea4395eea94869ae86aef3e7f72d1010100000000000...
  NTProofStr: b0eea4395eea94869ae86aef3e7f72d1
  Response Version: 1
  Hi Response Version: 1
  Z: 000000000000
  Time: Apr 18, 2019 14:17:09.242052800 UTC
  NTLMv2 Client Challenge: 26b00961558b7b4a
  Z: 00000000
  > Attribute: NetBIOS computer name: TEST-01
  > Attribute: NetBIOS domain name: PREEMPT
  > Attribute: DNS computer name: TEST-01.preempt
  > Attribute: DNS domain name: preempt
  > Attribute: DNS tree name: preempt
  > Attribute: Timestamp
  ▼ Attribute: Channel Bindings
    NTLMV2 Response Item Type: Channel Bindings (0x000a)
    NTLMV2 Response Item Length: 16
    Channel Bindings: 26b0b57ea3af385ae64834351e5a2f49
  > Attribute: Flags
  > Attribute: Restrictions
  ▼ Attribute: Channel Bindings
    NTLMV2 Response Item Type: Channel Bindings (0x000a)
    NTLMV2 Response Item Length: 16
    Channel Bindings: 00000000000000000000000000000000
  > Attribute: Target Name: HTTP/10.1.1.1
  > Attribute: End of list
```


NTLM: NEW VULNERABILITIES

- **EPA (Enhanced Protection for Authentication) bypass**
 - What if the NTLM_AUTHENTICATE message includes a MIC?
 - **DROP THE MIC!**

Original NTLM_AUTHENTICATE:

```
> NTLM Secure Service Provider
    NTLMSSP identifier: NTLMSSP
    NTLM Message Type: NTLMSSP_AUTH (0x00000003)
> Lan Manager Response: 0000000000000000000000000000000000000000000000000000000000000000
    LMv2 Client Challenge: 0000000000000000
> NTLM Response: b0eea4395eea94869ae86aef3e7f72d1010100000000000...
> Domain name: PREEMPT
> User name: user01
> Host name: TEST-01
> Session Key: f2ee625796ccac3fd657e015dd25454a
> Negotiate Flags: 0xe2888215, Negotiate 56, Negotiate Key Exchange, Negotiate
> Version 6.1 (Build 7601); NTLM Current Revision 15
    MIC: e746de89e1e239ad880738eccfe687dc
```



Modified NTLM_AUTHENTICATE:

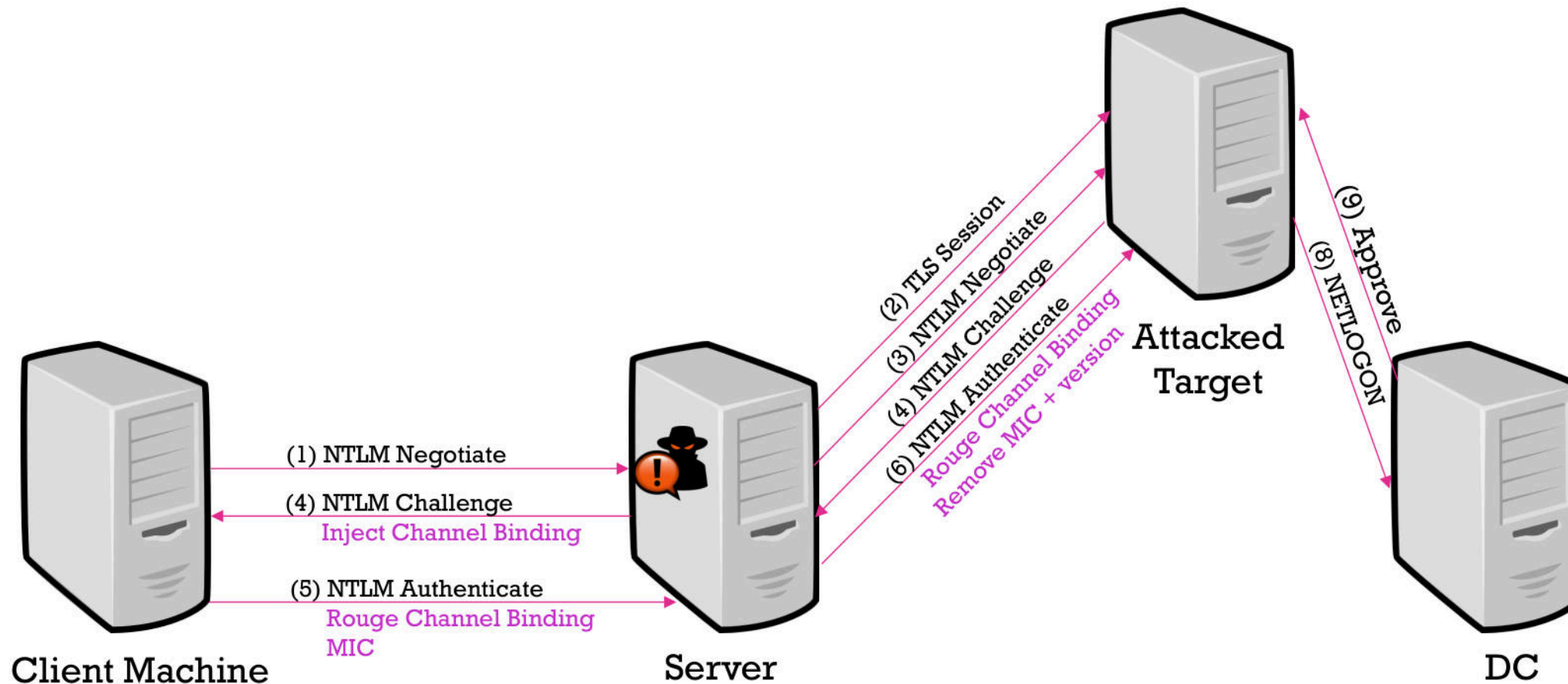
```

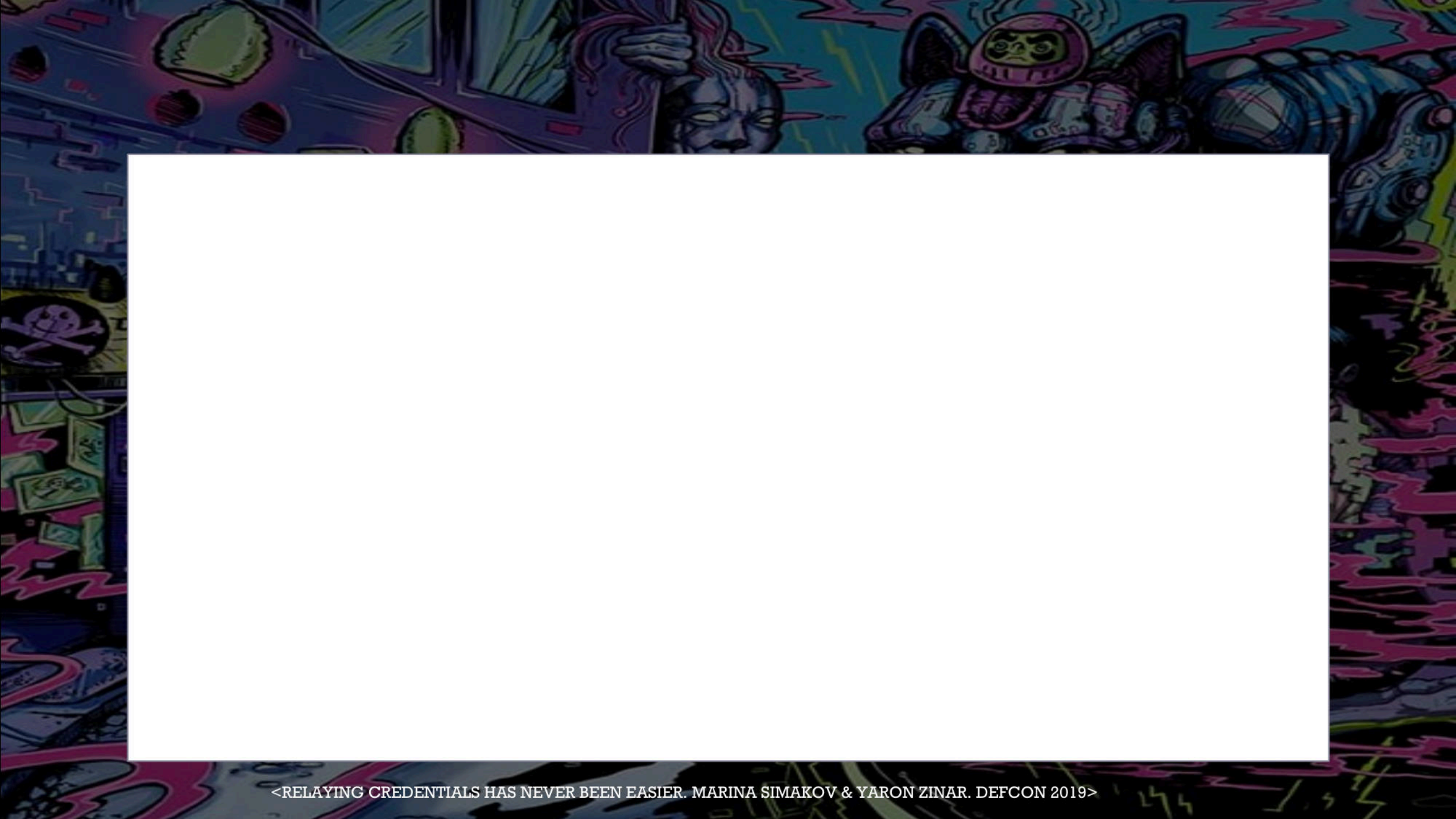
    NTLM Secure Service Provider
      NTLMSSP identifier: NTLMSSP
      NTLM Message Type: NTLMSSP_AUTH (0x00000003)
    > Lan Manager Response: 0000000000000000000000000000000000000000000000000000000000000000
      LMv2 Client Challenge: 0000000000000000
    > NTLM Response: b0eea4395eea94869ae86aef3e7f72d1010100000000000...
    > Domain name: PREEMPT
    > User name: user01
    > Host name: TEST-01
    > Session Key: 30002e0030002e003300390000000000
    > Negotiate Flags: 0xa0880205, Negotiate 56, Negotiate 128, Negotiate Target

```


NTLM: NEW VULNERABILITIES

- EPA (Enhanced Protection for Authentication) bypass





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NTLM: NEW VULNERABILITIES

- EPA bypass - Fix:

- Servers deny authentication requests which include more than one channel binding value

- Issues:

- Some clients don't support EPA & don't add a MIC (Firefox on Linux or MacOS)
- These clients are still vulnerable to the EPA bypass
- One such client is enough to make the entire domain vulnerable

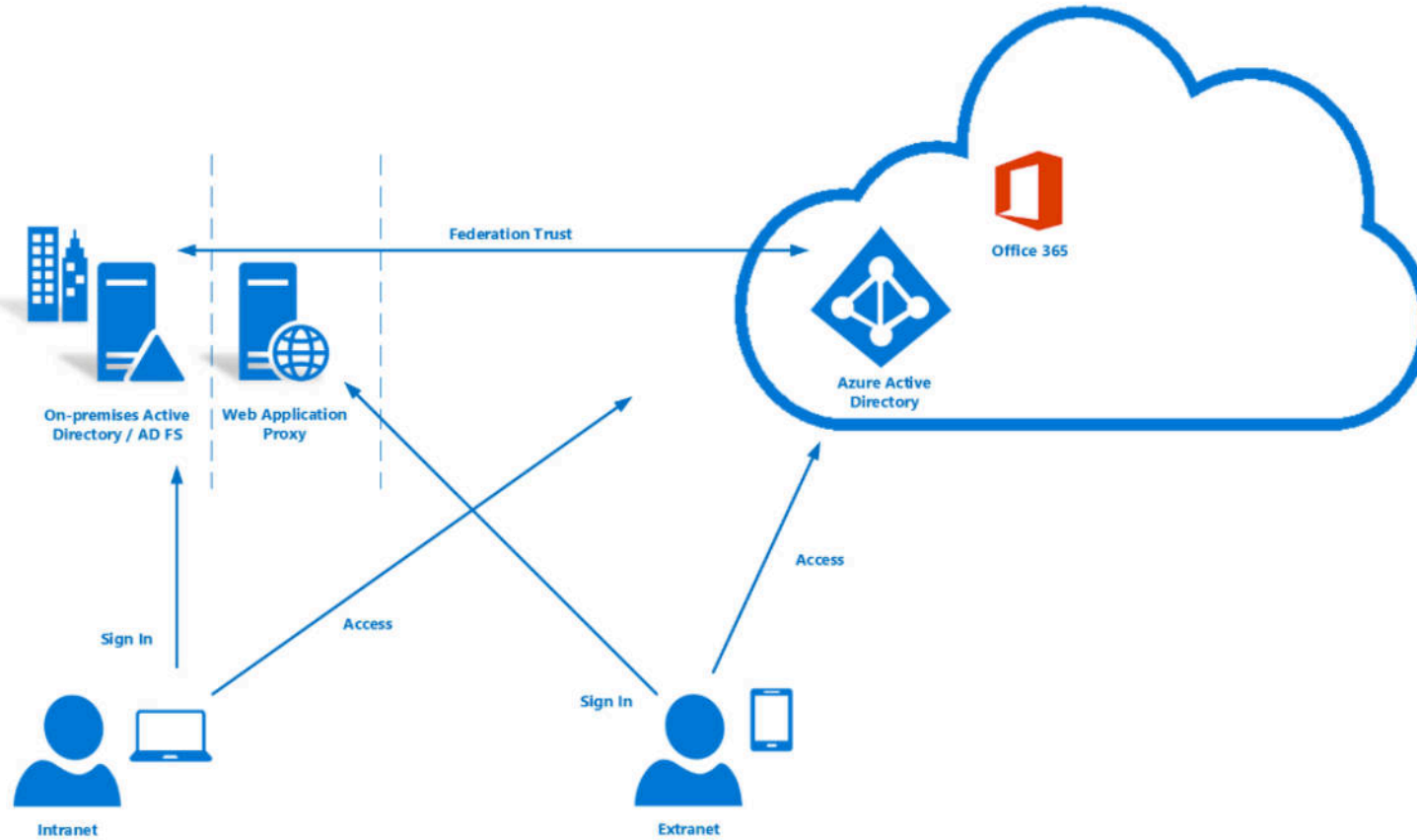


ATTACKING AD-FS

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ATTACKING AD-FS

■ AD-FS Architecture



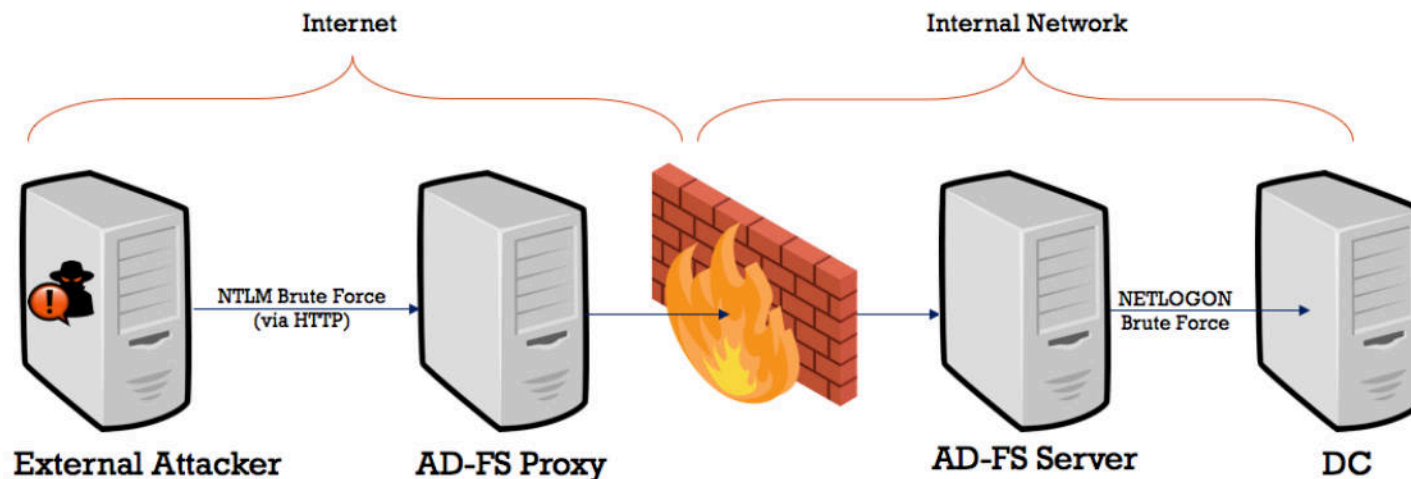
<https://www.sherweb.com/blog/office-365/active-directory-federation-services/>

ATTACKING AD-FS

- **AD-FS Proxy**
 - Open to the internet
 - Easy target for brute-force/password spraying attacks
- **External Lockout Policy**
 - Locks the user coming from the external network after exceeding the **Extranet Lockout Threshold**
 - Has effect when: **Extranet Lockout Threshold < AD Lockout Threshold**
 - Prevents brute-force-attacks
 - Prevents malicious account lockouts

ATTACKING AD-FS

- **WIA (Windows Integrated Authentication)**
 - Use Kerberos or NTLM SSO capabilities to authenticate to AD-FS
 - WIA authentications were accepted by the AD-FS proxy
 - **NTLM relay** against the AD-FS proxy from the external network
 - **NTLM authentications** target at the AD FS proxy allowed attackers to bypass the external lockout policy (CVE-2019-1126)



ATTACKING AD-FS

- **WIA (Windows Integrated Authentication)**
 - **Kerberos authentications** allowed attackers to brute-force the AD-FS service account's password
 - Generate service tickets using different passwords and send to AD-FS proxy
 - If password is successfully guessed -> log into cloud resources using any desired privileges
 - No logs generated for unsuccessful attempts
 - **Reverse-Kerberoasting!**



TAKEAWAYS

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TAKEAWAYS

- Patch all vulnerable machines!
- Restrict NTLM usage as much as possible
 - NTLM authentication is susceptible to NTLM relay attacks
 - Always prefer Kerberos usage
- Disable NTLMv1 in your environment
 - Configure the GPO 'Network security: LAN Manager authentication level' to: 'Send NTLMv2 response only. Refuse LM & NTLM'
 - <https://docs.microsoft.com/en-us/windows/security/threat-protection/security-policy-settings/network-security-lan-manager-authentication-level>
- Incorporate NTLM relay mitigations:
 - SMB & LDAP signing
 - LDAP channel binding
 - EPA

CREDITS

- The Preempt Research Team
 - Eyal Karni (@eyal_karni)
 - Sagi Sheinfeld
- Alberto Solino (@agsolino)
 - Some of the vulnerabilities are merged into impacket!
 - <https://github.com/SecureAuthCorp/impacket>

The image is a meme. It features a central photograph of a man with a grey beard and hair, wearing a dark pinstriped suit jacket over a light-colored shirt. He is sitting at a table with a green beer bottle in front of him. The background of the photo is dark and textured. The photo is framed by a white border. Above and below the photo are white text boxes with black text. The entire meme is set against a colorful, abstract background with blue, purple, and green swirling patterns and various symbols like a skull and a cat.

**I DON'T
ALWAYS USE NTLM**

**BUT WHEN I
DO, I GET RELAYED**



THANK YOU!

<RELAYING CREDENTIALS HAS NEVER BEEN EASIER. MARINA SIMAKOV & YARON ZINAR. DEFCON 2019>